

ALZHEIMER'S DISEASE PROGRESSION PREDICTION

Project Report

ADNI Dataset | Machine Learning for Early Detection

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Dataset	Alzheimer's Disease Neuroimaging Initiative (ADNI)
Participants	2,968 baseline participants
Goal	Predict progression to Alzheimer's disease before symptoms appear
Models	Logistic Regression, SVM, Random Forest, XGBoost, Neural Network
Data Type	Tabular — cognitive tests, MRI volumetrics, amyloid PET, genetics, demographics
Evaluation	AUC-ROC, F1-Score

1. Introduction

Alzheimer's disease (AD) is a progressive neurodegenerative disorder and the leading cause of dementia worldwide, responsible for approximately 70% of all dementia cases. It is characterized by the gradual accumulation of amyloid-beta plaques and neurofibrillary tau tangles in the brain — a pathological process that begins silently, often 15 to 20 years before any clinical symptom appears. By the time a patient receives a formal diagnosis, extensive and irreversible neuronal damage has already

occurred across critical brain regions including the hippocampus, entorhinal cortex, and prefrontal cortex.

The global burden of Alzheimer's disease is severe and growing. Approximately 55 million people worldwide currently live with dementia, with AD accounting for the majority of cases. This number is projected to nearly triple by 2050 due to aging populations. In addition to its devastating personal impact on patients and families, AD places an enormous economic burden on healthcare systems — estimated at over 1.3 trillion USD annually worldwide. Despite decades of research, there is currently no cure, and existing medications only manage symptoms without altering the disease trajectory.

The most critical window for intervention lies in the pre-symptomatic and early symptomatic stages of the disease. At the Mild Cognitive Impairment (MCI) stage, individuals exhibit measurable cognitive decline but retain functional independence. A significant proportion of MCI patients — particularly those with amnesic MCI — will progress to full Alzheimer's dementia within three to five years. Identifying which individuals will progress, and doing so years in advance, enables timely therapeutic intervention, clinical trial enrollment, and informed care planning.

Traditional diagnostic approaches rely on clinical assessment, neuropsychological testing, and manual neuroimaging interpretation — methods that are costly, subjective, and often delayed until symptoms are already advanced. Machine learning (ML) offers a fundamentally different approach: by learning patterns from large multimodal datasets combining cognitive test scores, brain MRI measurements, amyloid PET imaging, genetic markers, and demographic information, ML models can identify individuals at high risk of progression years before a clinical diagnosis would be made. This project applies machine learning to this early detection challenge using data from the Alzheimer's Disease Neuroimaging Initiative (ADNI).

2. Problem Statement

Can machine learning predict — at the time of a patient's baseline clinical visit — whether that patient will progress to a worse Alzheimer's diagnosis in the future, using non-invasive and clinically accessible measurements?

2.1 The Core Challenge

The fundamental challenge this project addresses is binary progression prediction: given a participant's data at their baseline visit (cognitive test scores, brain MRI measurements, amyloid PET results if available, APOE4 genetic status, and demographic information), predict whether that participant will

progress to a worse diagnosis — from Cognitively Unimpaired to MCI, or from MCI to Dementia — at any future follow-up visit.

This is a substantially harder problem than simple diagnosis classification (identifying who already has AD), because it requires the model to detect subtle early signals of future deterioration that may not yet be clinically visible. It is also the most clinically valuable version of the problem, as accurate early identification enables intervention before irreversible damage accumulates.

2.2 Why This Problem Is Difficult

- The progression signal is subtle — early-stage changes in brain volume, cognitive scores, and amyloid burden are small and overlapping between progressors and non-progressors at baseline.
- The dataset is severely class-imbalanced — only approximately 9.8% of baseline participants progress within the follow-up window, making standard accuracy a misleading metric.
- Data is multimodal and heterogeneous — cognitive tests, MRI measurements, PET imaging, and genetics exist on completely different scales and have different patterns of missingness.
- Not all participants have all modalities — amyloid PET was only collected for approximately 39% of participants, and MRI volumetrics for approximately 16%, requiring strategies to handle missingness without discarding data.
- Follow-up data is variable — participants have different numbers of follow-up visits over different time horizons, making the definition of 'progression' non-trivial.

2.3 The Gap in Existing Research

While numerous studies have applied machine learning to AD prediction, several important gaps remain in the published literature that this project directly addresses:

- Most published studies rely on a single ADNI phase, limiting their cohort to approximately 500–900 participants. This project uses all ADNI phases combined, yielding 2,968 baseline participants — a larger and more diverse cohort.
- Studies that achieve the highest AUC values (0.84–0.91) depend heavily on amyloid PET imaging, which is expensive and unavailable in most clinical settings. This project builds both a PET-free model for general clinical use and a PET-enriched model for specialized centers.
- Most studies focus exclusively on MCI patients. This project includes both Cognitively Unimpaired and MCI participants, enabling prediction at an even earlier stage.
- Very few studies provide a systematic comparison of multiple model families on the same dataset. This project trains and evaluates five models — Logistic Regression, SVM, Random Forest, XGBoost, and Neural Network — under identical conditions.

2.4 Project Objectives

This project has four primary objectives:

- Build a clean, multimodal, analysis-ready dataset from seven raw ADNI CSV files spanning all study phases, combining cognitive, imaging, genetic, and demographic data into a single baseline modeling dataset.
- Train and systematically compare five machine learning models — Logistic Regression, Support Vector Machine, Random Forest, XGBoost, and Neural Network — on the progression prediction task under identical preprocessing and evaluation conditions.
- Quantify the added predictive value of amyloid PET by comparing model performance with and without CENTILOIDS as a feature, directly addressing the question of whether expensive PET imaging is necessary for accurate progression prediction.

3. Dataset Details

3.1 Data Source

All data used in this project was sourced from the Alzheimer's Disease Neuroimaging Initiative (ADNI), a large-scale longitudinal multicenter study launched in 2004 and designed to identify and validate biomarkers for the early detection and tracking of Alzheimer's disease. ADNI collects serial MRI, PET, biological markers, genetics, and clinical and neuropsychological assessments from participants across more than 50 sites in North America. As of 2024, the ADNI database supports over 5,600 publications and is the definitive global benchmark for AD machine learning research.

The project uses data downloaded from the ADNI IDA (Image and Data Archive) portal. Seven raw CSV files were loaded, each corresponding to a distinct clinical domain. A central data dictionary file (DATADIC.csv) was included to assist in understanding column meanings across tables.

3.2 Raw Data Files

File	Domain	Description	Raw Columns
APOERES.csv	Genetics	APOE genotype data for each participant	~15
CDR.csv	Clinical Rating	Clinical Dementia Rating scale scores	~25
DXSUM.csv	Diagnosis	Visit-level diagnosis records; primary anchor for merging	~30
NEUROBAT.csv	Neuropsychology	Cognitive battery scores across multiple test domains	~80
PTDEMOG.csv	Demographics	Participant demographics: gender, DOB, education, race	~40
UCBERKELEY_AMY_6MM.csv	Amyloid PET	Amyloid PET data including Centiloid values and amyloid status	~25

File	Domain	Description	Raw Columns
UCSFFSX7.csv	MRI Volumetrics	Structural MRI brain region volume measurements (FreeSurfer)	~400+

3.3 Cohort Definition

The final modeling cohort consists of 2,968 participants at their baseline visit (first clinical assessment). The cohort was restricted to participants with a baseline diagnosis of Cognitively Unimpaired (CU, DIAGNOSIS=1) or Mild Cognitive Impairment (MCI, DIAGNOSIS=2). Participants already diagnosed with Dementia at baseline were excluded, as the project's goal is to predict future progression to dementia — not classify those already diagnosed. The cohort breakdown is as follows:

Group	Count	Percentage	Description
Cognitively Unimpaired (CU)	1,516	51.1%	No cognitive impairment at baseline
Mild Cognitive Impairment (MCI)	1,452	48.9%	Cognitive decline present but functionally independent
Total	2,968	100%	Baseline modeling cohort

3.4 Target Variable — PROGRESSOR

The target variable PROGRESSOR was constructed from longitudinal follow-up data. For each participant, the worst diagnosis observed at any follow-up visit (MONTH > 0) was compared to their baseline diagnosis. If any future visit showed diagnostic worsening — CU progressing to MCI or Dementia, or MCI progressing to Dementia — the participant was labeled as a progressor (PROGRESSOR = 1). Participants with stable or improving trajectories were labeled as non-progressors (PROGRESSOR = 0).

Label	Count	Percentage	Meaning
Non-Progressor (0)	2,677	90.2%	Diagnosis stable or improved across all follow-up visits
Progressor (1)	291	9.8%	Diagnosed with worse condition at any future visit
Total	2,968	100%	

The class imbalance (90.2% vs 9.8%) is a defining characteristic of this dataset and requires explicit handling during model training. Standard accuracy is therefore not used as an evaluation metric. AUC-ROC and F1-score are the primary metrics, consistent with all published studies on this task.

3.5 Feature Set

The final modeling dataset contains 17 features plus the target variable, organized across five clinical domains. Total dimensionality was reduced from over 590 raw columns across all source files to 17 meaningful features through domain-informed column selection, feature engineering, and normalization.

Feature	Domain	Description	Coverage
AGE_AT_VISIT	Demographics	Age at baseline visit, derived from date of birth	99.9%
PTGENDER	Demographics	Biological sex (1=Male, 2=Female)	100%
PTEDUCAT	Demographics	Years of education — cognitive reserve proxy	99.9%
APOE4_COUNT	Genetics	Number of APOE ε4 alleles (0, 1, or 2)	75.4%
LIMMTOTAL	Cognition	Logical Memory Immediate Recall score	99.6%
LDELTOTAL	Cognition	Logical Memory Delayed Recall score	99.5%
AVTOT4	Cognition	RAVLT Trial 4 — verbal learning efficiency	99%+
AVDEL30MIN	Cognition	RAVLT 30-min Delayed Recall — hippocampal integrity	99%+
AVDELTOT	Cognition	RAVLT Recognition Score — encoding efficiency	99%+
TRAASCOR	Cognition	Trail Making Test A — processing speed	99%+
TRABSCOR	Cognition	Trail Making Test B — executive function	99%+
CATANIMSC	Cognition	Category Fluency Animals — semantic memory	99%+
CATVEGESC	Cognition	Category Fluency Vegetables — semantic memory	99%+
CENTILOIDS	Amyloid PET	Continuous amyloid burden measure from PET imaging	39.3%
HIPPOCAMPUS_NORM	MRI	Bilateral hippocampal volume normalized by ICV	15.8%
ENTORHINAL_NORM	MRI	Bilateral entorhinal cortex volume normalized by ICV	15.7%
VENTRICLES_NORM	MRI	Bilateral lateral ventricle volume normalized by ICV	15.7%

3.6 Missing Data Strategy

Missing data was handled differently depending on the nature of each feature. For non-imaging features (cognitive tests, demographics, and APOE4), Multiple Imputation by Chained Equations

(MICE) was applied using scikit-learn's `IterativeImputer` with 10 iterations. MICE is preferred over simpler strategies because it models each feature's missing values using all other features, preserving correlation structure rather than introducing bias toward the mean.

For imaging features (HIPPOCAMPUS_NORM, ENTORHINAL_NORM, VENTRICLES_NORM, and CENTILOIDS), deliberate non-imputation was applied. These features are missing because the scans were simply not collected for those participants — imputing them would mean fabricating imaging biomarkers for patients who never underwent MRI or PET scanning, which would be clinically misleading. These features remain as NaN for participants without imaging data, and the modeling strategy accounts for this through separate model variants.

3.7 Key Preprocessing Steps

- VISCODE standardization: All visit codes across ADNI phases mapped to a unified numeric MONTH field representing months since baseline.
- Nearest-neighbor temporal merge: MRI and PET scans matched to clinical visits within a 6-month tolerance window, maximizing imaging data utilization.
- Smart MRI QC filter: 'Pass' and region-specific 'Partial' scans retained; only outright 'Fail' scans excluded.
- ICV normalization: All MRI volumes normalized by intracranial volume to remove brain size confound.
- APOE4_COUNT engineering: Raw genotype strings (e.g., '3/4') converted to integer allele counts (0, 1, 2).
- AGE_AT_VISIT engineering: Derived from date of birth and visit date; raw date columns dropped.

4. Literature Review

4.1 Introduction

Machine learning has emerged as a transformative tool for early Alzheimer's disease detection over the past decade. By learning patterns from multimodal clinical data — cognitive assessments, brain imaging, genetic markers, and demographics — ML models can identify individuals at risk of progression years before a clinical diagnosis. This section reviews the most relevant published research, focusing on the datasets used, models applied, performance benchmarks achieved, and the gaps that the current project addresses.

4.2 The ADNI Dataset as the Research Benchmark

The Alzheimer's Disease Neuroimaging Initiative (ADNI) is the most widely used dataset in AD machine learning research globally. As of 2024, ADNI supports over 5,600 publications and contains longitudinal data from over 2,000 participants collected across multiple phases since 2004. Its standardized

multimodal data collection — combining MRI, PET, cognitive tests, genetics, and clinical assessments — makes it the definitive benchmark for AD ML research. The current project uses all ADNI phases combined, yielding 2,968 baseline participants, which is larger than most published studies that typically use a single phase with 500–900 participants.

4.3 The Five Models in the Literature

Logistic Regression

Logistic Regression (LR) is the most commonly used baseline model in AD prediction research due to its full interpretability and clinical familiarity. Across published studies using ADNI and OASIS datasets, LR consistently achieved accuracies between 0.74 and 0.80 and AUC values around 0.70–0.75 on AD progression tasks. While insufficient as a standalone clinical tool, it establishes the performance floor that more complex models must exceed and provides a transparent reference for feature coefficient analysis. Every major comparative study in the literature includes LR as the baseline model.

Support Vector Machine (SVM)

Support Vector Machines have been among the most widely validated models in AD classification literature. Across multiple independent studies using ADNI and OASIS datasets, SVM demonstrated strong performance, reaching an accuracy of 0.9677 in one study and 0.92 with a precision of 0.919 in another. A 2024 study by Alatrany et al. using National Alzheimer's Coordinating Center data (169,408 records, 1,024 features) found that SVM achieved an F1 score of 98.9% for binary AD classification and 90.7% for multiclass CN/MCI/AD classification. SVM's effectiveness on medical tabular data comes from its ability to find optimal decision boundaries in high-dimensional feature spaces, making it particularly well-suited for clinical biomarker datasets.

Random Forest

Random Forest (RF) consistently ranks among the top-performing models across AD prediction literature. Its key advantages are robustness to missing values, resistance to overfitting through ensemble averaging, native feature importance rankings, and strong performance without extensive hyperparameter tuning. Across multiple studies using ADNI and OASIS, RF achieved accuracies above 0.84. In a 2024 study evaluating eight ML models for MCI-to-AD progression prediction on ADNI, Random Forest ranked second with an AUC of 0.83. A survival analysis study using Random Forest on ADNI and AIBL datasets achieved a concordance index of 0.84, confirming its consistency across different task formulations and dataset configurations.

XGBoost

XGBoost has emerged as the leading model for tabular ADNI data in recent literature. In the 2024 study most directly comparable to the current project — evaluating eight ML models for MCI-to-AD progression on ADNI — XGBoost achieved the highest AUC of 0.84, outperforming all alternatives including KNN, LR, SVM, Decision Trees, RF, Gradient Boosting, and ANNs.

Neural Network (MLP)

Neural networks have demonstrated strong performance on AD prediction tasks, particularly when multiple data modalities are combined. A study using ADNI data combined 3D CNNs for MRI imaging with stacked denoising autoencoders for clinical and genetic data, showing that deep models

outperformed all shallow models including SVM, RF, and KNN on the multimodal classification task. A 2025 study used Long Short-Term Memory networks for temporal dependencies in longitudinal data alongside a feedforward neural network for static features, demonstrating improved accuracy over cross-sectional approaches. An ensemble of five deep learning architectures achieved 99.5% accuracy on ADNI MRI scans. For the tabular dataset in the current project, a Multilayer Perceptron (MLP) is the appropriate neural network architecture — it takes rows of numbers as input and captures non-linear feature interactions that linear and tree-based models may miss, without requiring image data.

4.4 Handling Class Imbalance

Class imbalance is a universal challenge in AD progression studies. Published studies address it through SMOTE (Synthetic Minority Over-sampling Technique), which generates synthetic minority class samples, and class weighting, which penalizes models more heavily for misclassifying progressors. A direct comparison of four sampling strategies found NEAR-MISS undersampling most effective when combined with XGBoost. Because accuracy is meaningless under severe class imbalance, all reviewed studies use AUC-ROC and F1-score as primary metrics — an approach the current project follows.

4.5 Most Predictive Features Identified

Across the reviewed literature, a consistent set of features emerges as most predictive of AD progression: RAVLT delayed recall and Logical Memory delayed recall as the strongest cognitive markers of hippocampal decline; hippocampal and entorhinal cortex volumes as the most informative structural MRI features; amyloid Centiloids as the most direct pathological biomarker; APOE4 allele count as the strongest genetic risk factor; and age and education as the most important demographic covariates. These findings directly informed the current project's feature selection strategy.

4.6 Summary of Key Studies

Study / Year	Dataset	Best Model	Performance	Task
Springer Nature, 2024	ADNI	XGBoost	AUC 0.84	MCI → AD Progression
Sheng et al., 2023	ADNI + NACC	XGBoost + SHAP	AUC 0.91	CN/MCI/AD Classification
Alatrany et al., 2024	NACC + ADNI	SVM	F1 98.9%	Binary NC vs AD
Sci. Reports, 2025	ADNI + AIBL	Random Forest	C-index 0.84	CN → MCI Survival
Triage Framework, 2024	ADNI (MCI)	Interpretable ML	AUROC 0.929	MCI → AD
Li et al., 2024	ADNI	XGBoost, RF, SVM	Competitive AUC	Early Detection
BMC Med Res, 2025	ADNI (MCI)	Gradient Boosting	C-index comparison	MCI → AD Survival

4.7 How This Project Advances the Literature

- Largest multimodal ADNI cohort: 2,968 participants combining all phases, versus 500–900 in most published studies.
- Two-model clinical strategy: A PET-free model for general clinics and a PET-enriched model for specialized centers, directly quantifying the added value of amyloid imaging.
- Both CU and MCI participants: Enables prediction at the earliest possible stage, before MCI is even diagnosed.
- Five-model systematic comparison: LR, SVM, RF, XGBoost, and Neural Network evaluated under identical conditions — a rigorous benchmark most papers do not provide.

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